

PROJECT PRESENTATION

ContextGraph

A Cross-AI Personal Context Engine

A structured context graph that stores identity, stack, and project knowledge once, so any MCP-compatible AI can read and update it through a personal API key.

Prepared 2026-09-09

Next.js · Supabase · Better Auth · Model Context Protocol



Project Overview



The Problem

Every AI session starts from zero. Context is siloed per platform, non-portable, and flat, with no scope containment between clients or projects.

Siloed

Non-portable

Flat



The Solution

A structured, hierarchical context graph, owned by the user, reachable by any MCP-compatible AI through a personal API key, and visualized in a web dashboard.

Hierarchical

Portable

Explicit

WHO IT'S FOR

● Primary

Developers and AI power users running multiple tools across multiple projects

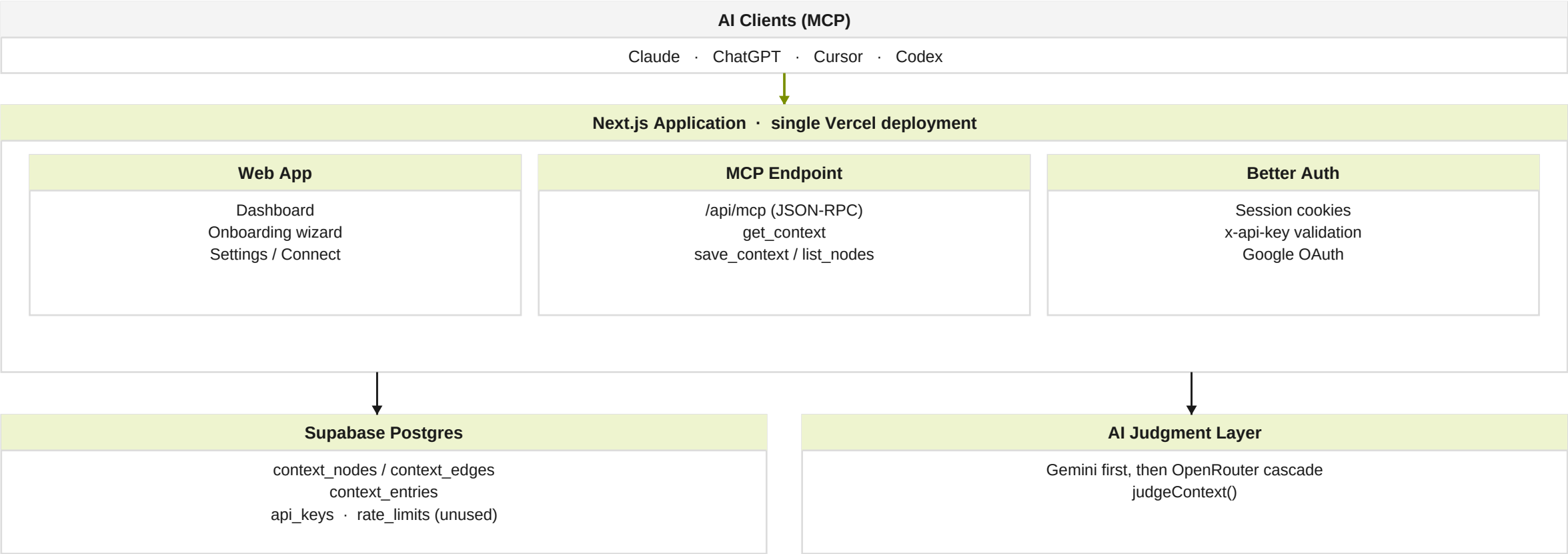
● Secondary

Freelancers and agency owners managing separate context per client

STACK

Next.js (App Router) · Supabase Postgres · Better Auth · react-force-graph-3d · Gemini / OpenRouter · Model Context Protocol

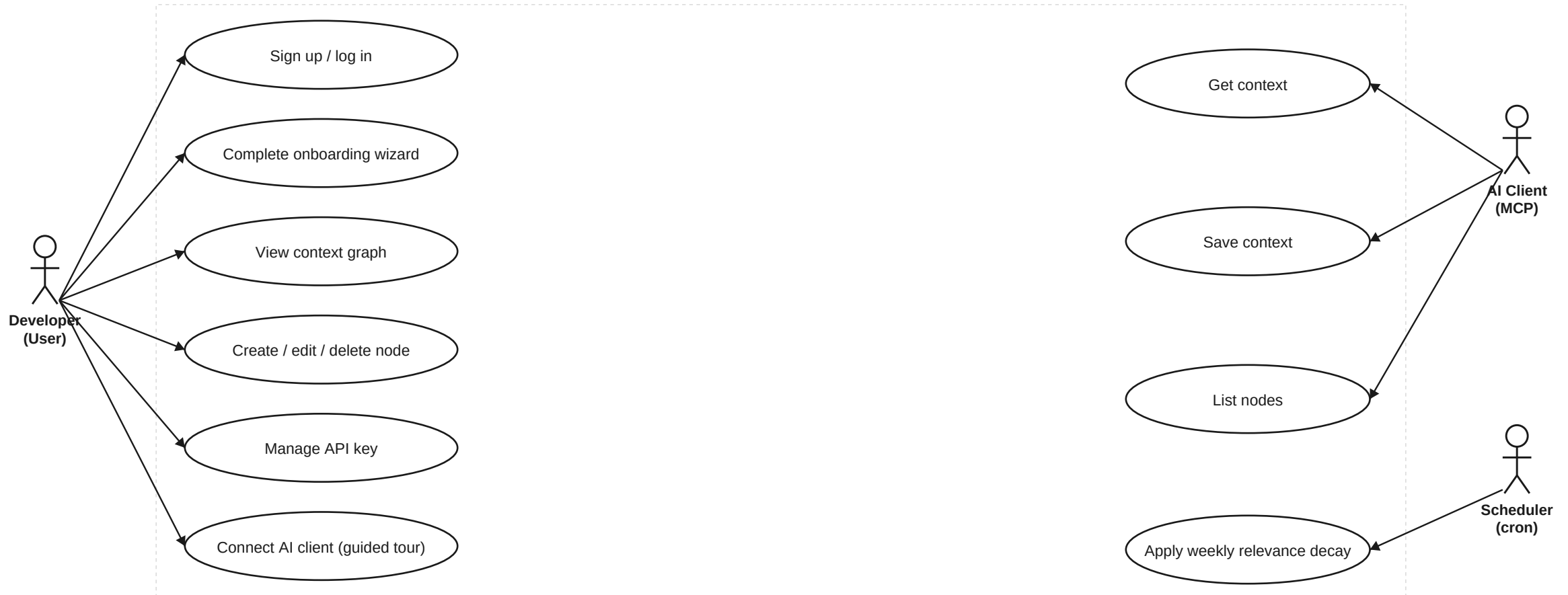
System Architecture



One Vercel deployment serves the marketing site, the authenticated app, and the MCP endpoint. There is no separate backend service.

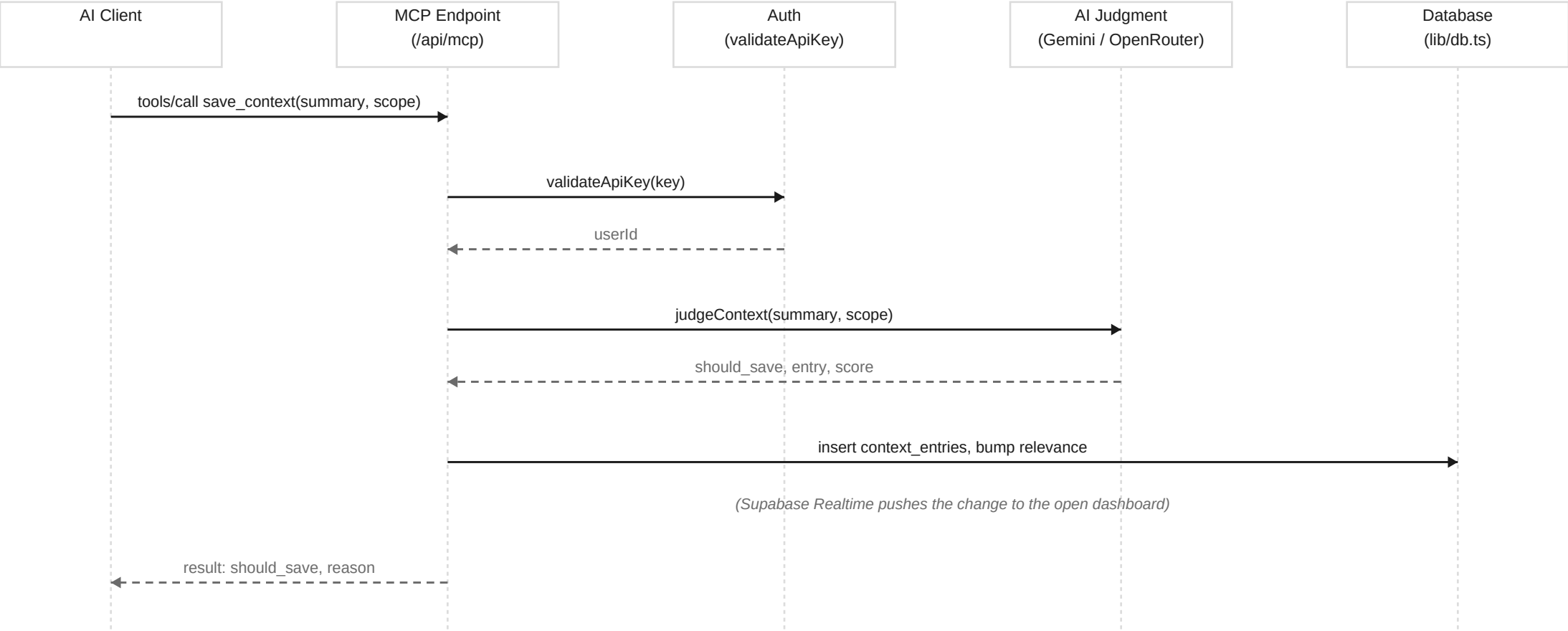
Use Case Diagram

ContextGraph



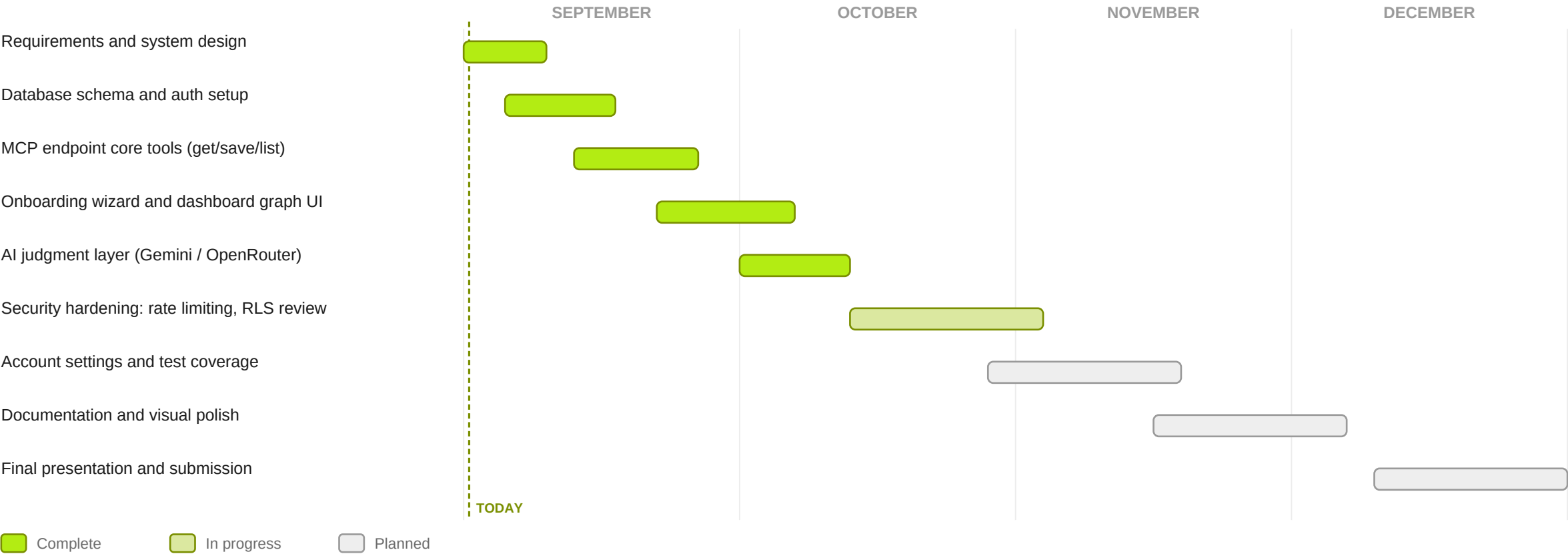
The AI Client actor represents any MCP-compatible tool authenticating with a personal API key.

Sequence: Save Context via MCP



If `should_save` is false, the AI client receives the reason and no write occurs. The judgment step, not the endpoint, decides what is worth keeping.

Project Timeline



Task Breakdown & Definition of Done

Module	Status	Notes
Auth (email/password + Google)	Done	Better Auth, proxy.ts route guard
Onboarding wizard	Done	Structured 5-step form, AI-authored node content
Dashboard graph view	Done	react-force-graph-3d, realtime updates
MCP endpoint (3 tools)	Done	JSON-RPC 2.0, API-key auth
Weekly relevance decay	Done	Cron route
Account settings (edit/delete)	Pending	UI exists, not yet wired to a route
MCP rate limiting	Pending	Table exists, not enforced
RLS / anon-key review	Pending	Needs verification against the live Supabase project

DEFINITION OF DONE

- ✓

Sign up and complete onboarding
- ✓

See the context graph in the dashboard
- ✓

Connect any MCP-compatible AI with a personal API key
- ✓

get_context loads context automatically
- ✓

/save adds a new entry to the dashboard graph

Risks & What's Next

KEY RISKS



No enforced rate limiting on the MCP endpoint

Wire up the existing rate_limits table before opening the endpoint to more clients.



Tenant isolation is fully programmatic, no RLS

Audit every lib/db.ts call for a userId filter and verify anon-key exposure on the live project.



Two parallel hierarchy fields can drift apart

Update parent_scope and context_edges together on every write, or collapse to one representation.



AI judgment depends on third-party model uptime

A Gemini-first, OpenRouter-fallback cascade already retries across models on transient errors.

WHAT'S NEXT



AI memory export parsing, turning a ChatGPT or Claude memory export into a first-class import.



Multiple API keys with labels and scoped permissions, such as a read-only key for a docs bot.



Node version history, so a past state of a context node can be reviewed or restored.



Public read-only graph sharing, for handing one project's context to a client.



A browser extension that auto-injects active context into web-based chat UIs.